

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY

9744/04

Paper 4 Practical

24 August 2017
Thursday

Candidates answer on the Question Paper.

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	55

Answer **all** the questions.

- 1** The enzyme lipase catalyses the hydrolysis of triglycerides into fatty acids and glycerol.

You are required to investigate the effect on the lipase-catalysed reaction of the independent variables:

- enzyme concentration
- presence of calcium ions

The substrate for lipase will be the triglycerides present in milk.

The progress of this hydrolysis can be monitored by using an indicator, **T**, which changes colour due to the production of fatty acids.

You are provided with the following solutions;

25cm³ of milk containing calcium ions, labelled **M+C**,

25cm³ of milk without calcium ions, labelled **M**,

25cm³ of indicator solution, labelled **T**,

30cm³ of sodium carbonate solution, labelled **A**,

20cm³ of 10% lipase solution, labelled **E10**,

20cm³ of 5% lipase solution, labelled **E5**,

Lipase is an irritant. You are advised to wear the eye protection provided. Contact of the solution with your skin should be avoided. If it touches your skin, wash it off with tap water.

Proceed as follows.

(a) Stage 1

Use the beaker or container provided to make a water-bath with warm water, between 38°C and 42°C.

Stage 2

Label four boiling tubes, **B1**, **B2**, **B3** and **B4**.

Using the syringes, put:

- 2cm³ of solution **M+C** into each of the boiling tubes labelled **B1** and **B2**,
- 2cm³ of solution **M** into each of the boiling tubes labelled **B3** and **B4**,
- 2cm³ of solution **T** into each of the boiling tubes labelled **B1**, **B2**, **B3** and **B4** and gently shake,
- 3cm³ of solution **A** into each of the boiling tubes labelled **B1**, **B2**, **B3** and **B4** and gently shake so that all the mixture turns blue. Minor variations in colour between the tubes can be ignored as long as the contents are blue.

Put the four boiling tubes into the water-bath for at least three minutes, before progressing to **Stage 3**.

Stage 3

After the boiling tubes have been in the water-bath for three minutes, start a stopwatch, which will be left running continuously throughout the investigation. Start and end times will be taken from this stopwatch.

Stage 4

Remove the boiling tubes labelled **B1** and **B3** from the water-bath and put them in a boiling tube rack. Use a syringe to put 2cm³ of solution **E10** into each of these two boiling tubes and mix well. Record the start times in Table 1.1, below.

Stage 5

Observe the boiling tubes **B1** and **B3** and record the times at which the colour changes (end times) in Table 1.1. If the time taken for the colour to change for any boiling tube is longer than five minutes, record 'no change'.

Stage 6

Remove the boiling tubes labelled **B2** and **B4** from the water-bath and put them in a boiling tube rack. Use a syringe to put 2cm³ of solution **E5** into each of these two boiling tubes and mix well. Record the start times in Table 1.1.

Stage 7

Observe the boiling tubes **B2** and **B4** and record the times at which the colour changes (end times) in Table 1.1. If the time taken for the colour to change for any boiling tube is longer than five minutes, record 'no change'.

Stage 8

Calculate the time taken for the colour to change for each of the boiling tubes **B1-B4**, and record this in Table 1.1. If the time taken for the colour to change for any boiling tube is longer than five minutes, record 'no change'.

Table 1.1

	B1	B2	B3	B4
Start time/s				
Time at which colour changes (end time)/s				
Time taken for colour to change/s				

[3]

- (b)** Prepare a table in the space below to show the effect of enzyme concentration and presence of calcium ions on the hydrolysis of triglycerides in milk. [6]

- (c)** Describe how you could set up a control for the effect of lipase on triglycerides.

[1]

- (d) Identify **one** significant source of error in measuring the dependent variable in this investigation.

[1]

- (e) State **two** ways in which the experimental procedure could be improved.

[2]

- (f) Explain why the method used in this investigation is not suitable for investigating the effect of pH on the activity of lipase.

[2]

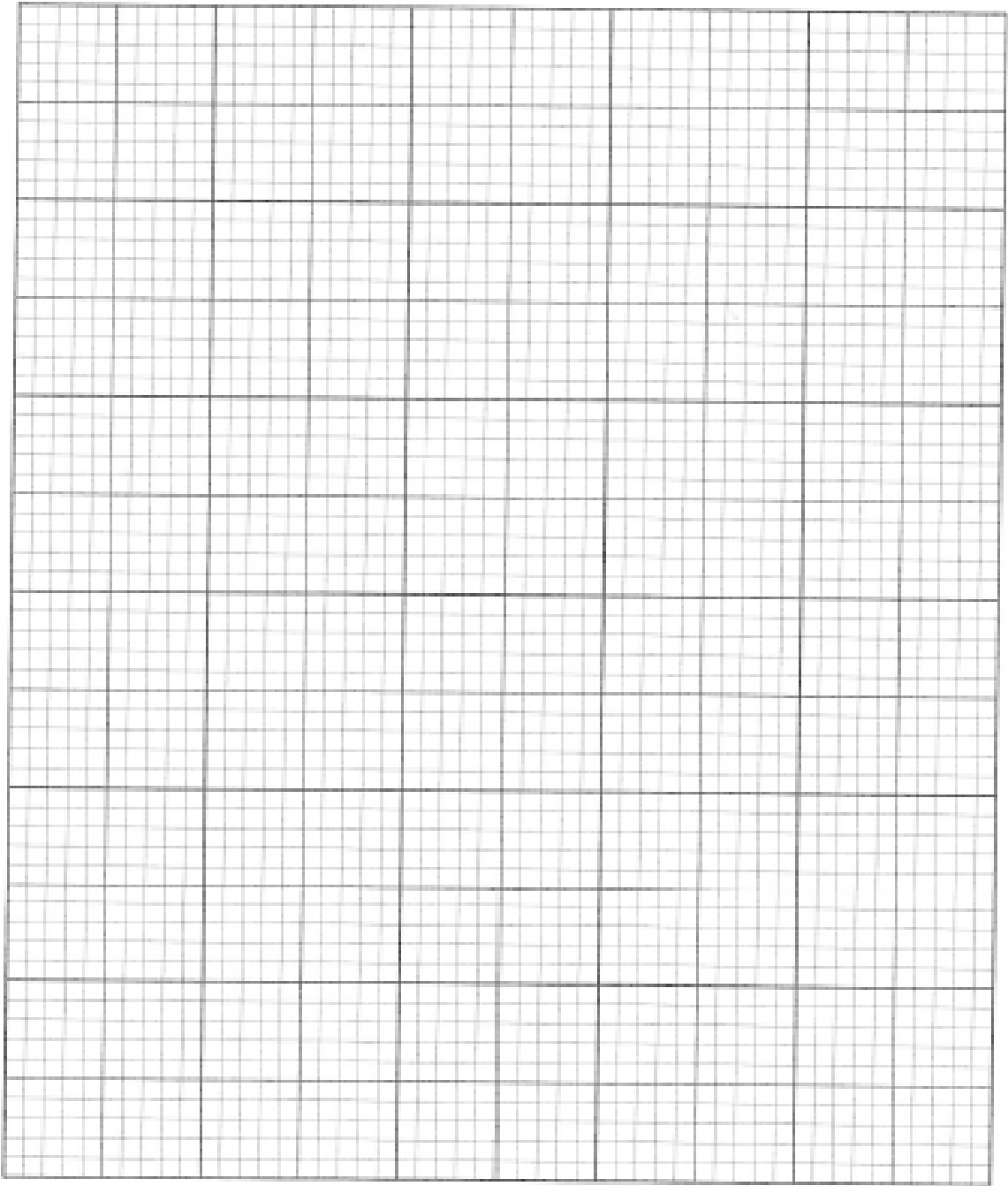
Some students carried out an investigation using lipase and found that its activity was affected by the concentration of copper sulfate solution. All other variables were kept constant.

The results of their investigation are shown in Table 1.2.

Table 1.2

Copper sulfate concentration/ $\times 10^{-3} \text{ mol dm}^{-3}$	Lipase activity/arbitrary units
1.0	26.0
2.0	12.0
3.0	5.0
4.0	2.5
5.0	1.5

- (g) Plot a graph of the data shown in Table 1.2, on the grid on the next page.



[4]

(h) Describe and explain these results.

[2]

[Total: 21]

- 2** **K1** and **K2** are stained, transverse sections of leaves from two different species of plant.
- (a)** **(i)** Make a large, labelled, plan drawing of **K1** to show the distribution of tissues in the leaf lamina (avoiding the midrib). Details of individual cells are **not** required.

[3]

- (ii)** Make a labelled, high-power drawing to show the detailed structure of **three** adjacent cells from the palisade mesophyll layer.

[3]

- (iii) Use the stage micrometer to determine the area of the field of view under high power. Calculate the **average** density of palisade mesophyll cells. **State the magnification used and show your working.**

Magnification used: _____

Average density of palisade mesophyll cells: _____ [3]

- (iv) Calibrate the eyepiece graticule using the stage micrometer so that you can use it to measure the length along one palisade mesophyll cell under a suitable magnification. Repeat until you have three measurements. **State the magnification used and show your working in calculating the average length.**

Magnification used: _____

Average length of palisade mesophyll cell: _____ [3]

- (b) The plant species from which **K2** was taken grows in a dry habitat. Examine **K2**, using your microscope. State **four** observable features that distinguish **K2** from **K1** and present the differences in a suitable format.

(c) You are required to investigate some aspects of the water.

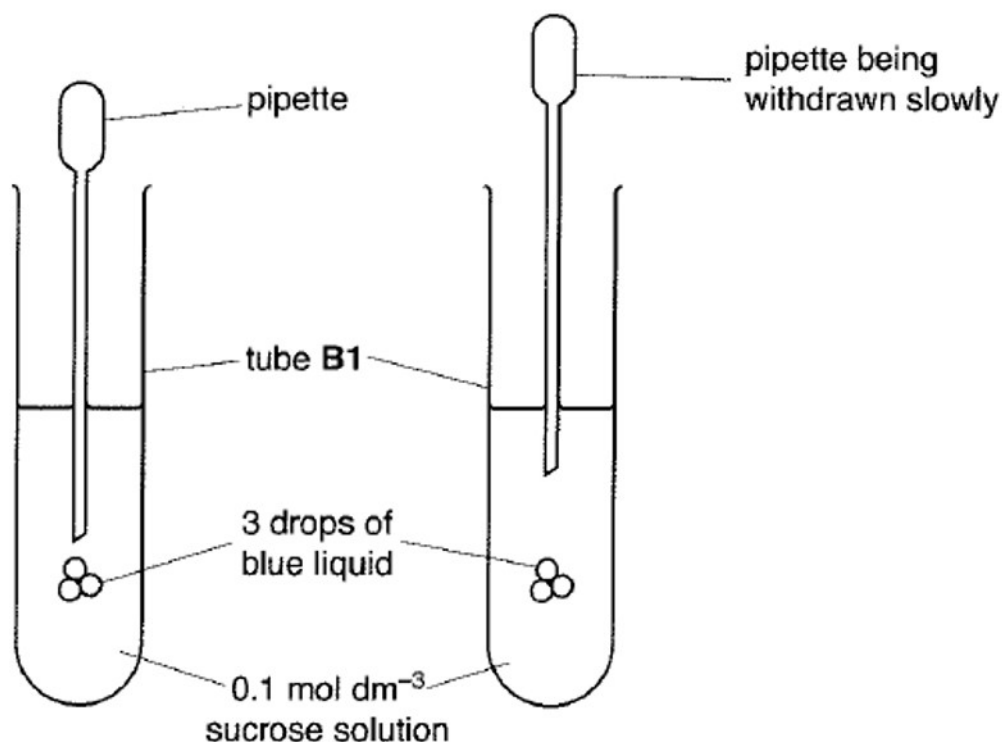
(i) Label four test tubes **B1**, **B5**, **B10** and **C5** respectively.

Using a syringe or pipette, prepare the three concentrations of sucrose using the water and 1.0 mol dm^{-3} sucrose solution provided. Record the volume of water and 1.0 mol dm^{-3} sucrose solution used in the table below. The total volume of different concentrations of sucrose should be 10 cm^3 . Then place 10 cm^3 of sucrose solution into each of these tubes.

Tube	B1	B5	B10
Concentration of sucrose solution/ mol dm^{-3}	0.1	0.5	1.0
volume of water/ cm^3			
volume of 1.0 mol dm^{-3} sucrose solution/ cm^3			

[2]

Transfer all 10 cm^3 0.5 mol dm^{-3} sucrose solution from **B5** into a test tube labelled **C5** and add three drops of the dye, methylene blue (labelled **MB**) to it. Shake tube **C5** to make the colour uniform. Suck up a little of this blue 0.5 mol dm^{-3} sucrose solution into a pipette and then, with the tip of this pipette held stationary, half way down the solution in tube **B1** (see diagram below), **very gently** release three drops from the pipette. **Do not squirt these drops into the solution. Withdraw the pipette slowly.**



Carefully observe the movement of the blue liquid. Repeat the procedure for **B10**.

- (i) Record your observations in a suitable format for **B1** and **B10**.

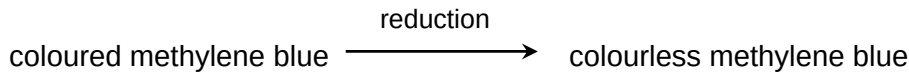
[2]

[Total: 20]

3 Planning question

Yeast undergoes aerobic respiration, breaking down glucose into carbon dioxide and water. This process is catalyzed by enzymes.

Methylene blue is an artificial **hydrogen acceptor** which is blue in the oxidised form and colourless when reduced.



A colorimeter can be used to measure the absorbance of light at 550 nm by methylene blue solution. When methylene blue is reduced and becomes colourless, its absorbance reading will become 0 arbitrary units. Yeast suspension has an absorbance value of 15 arbitrary units. The colorimeter is shown in **Fig. 4.1** below.

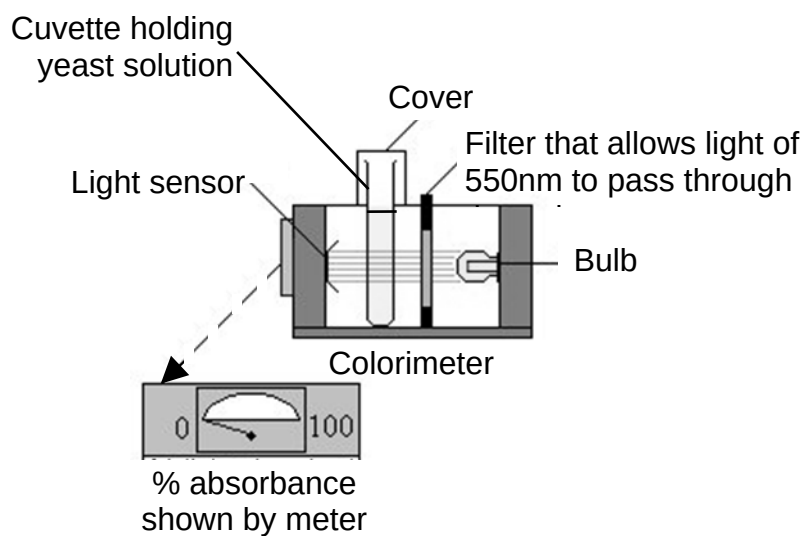


Fig. 4.1

Using this information and your own knowledge, design an experiment to test the hypothesis that:

“The rate of respiration in yeast cells is dependent on the concentration of glucose solution.”

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- identify independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- active yeast suspension
- 1.0 mol dm^{-3} glucose solution
- methylene blue
- test tubes
- stopwatch
- colorimeter
- 2 cuvettes
- beakers
- a variety of different sized measuring cylinders, syringes and pipettes for measuring volumes

[14]

[Turn over]

